Carnegie Mellon University

Aayan Sharma <aayans@andrew.cmu.edu>

Inquiry About Undergraduate Research Opportunities (Summer)

40 messages

Aayan Sharma <aayans@andrew.cmu.edu> Sun, Feb 22, 2026 at 10:35 AM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Dear Prof. Low,

I hope you are well. My name is Aayan Sharma, and I am a first year Tepper student with interests in ECE. I'm reaching out to see if you or any of your colleagues are looking for undergraduate research assistants for the summer.

I'm broadly interested in high-performance computing and systems work, especially performance optimization, hardware/software tradeoffs, and building efficient tools that make complex computing more practical and accessible. I've looked into your work and am particularly intrigued by your recent paper, "Architecture-Aware Models of AI Engines for High-Performance Matrix Matrix Multiplication," because I'm really interested in the blend of computer architecture and software computation needed to get real performance.

Additionally, outside of coursework I run GigaGeekTech, where I build workflow-specific workstation PCs for professionals (e.g., 3D modeling, rendering, photogrammetry, high frequency trading, ML, etc). Feel free to take a look at my work here (youtube.com/gigageek). That experience has trained me to think carefully about bottlenecks (compute vs memory vs storage), make data-driven tradeoffs, and make hardware decisions for workloads that require heavy computation. I aspire to build a career in computer hardware, so getting involved in your research would be a valuable learning opportunity and strongly aligned with my long-term interests.

If there are any opportunities available that I might be a good fit for, please let me know. I'd be happy to set up a brief meeting to discuss your work and where I could contribute. Thank you for your time and consideration!

Sincerely,
Aayan Sharma

Tze Meng Low <lowt@andrew.cmu.edu> Mon, Feb 23, 2026 at 11:32 AM
To: Aayan Sharma <aayans@andrew.cmu.edu>

Dear Aayan,

I'm happy to talk to see if there's a fit for us over summer. Would you be able to meet on Friday morning at about 11am in my office (HH A303?

Best regards,
Tze Meng
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Mon, Feb 23, 2026 at 3:59 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Thank you for your reply. Unfortunately, I have class at 11AM on Friday. I am available any time after 1PM Friday. Let me know what time works for you best.

Thanks again,
Aayan Sharma
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Thu, Feb 26, 2026 at 12:41 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Dear Prof. Low,

Good afternoon, I hope you are doing well.

I wanted to follow up on scheduling a time to meet. I'm still unavailable at 11:00am on Friday due to class, but I'm available anytime after 12:00pm on Friday.

Thank you,
Aayan Sharma

[Quoted text hidden]

Tze Meng Low <lowt@andrew.cmu.edu> Thu, Feb 26, 2026 at 1:31 PM
To: Aayan Sharma <aayans@andrew.cmu.edu>

Hi Aayan,

How about 3:30-4pm?

Best regards,
Tze Meng
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Thu, Feb 26, 2026 at 4:01 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Yes, that works for me. See you then. Thank you!
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Mon, Mar 2, 2026 at 10:08 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Dear Professor Low,

Thank you very much for taking the time to meet with me this past Friday. I really enjoyed our conversation and appreciated your guidance.

Since we met, I've read "FATHOM: Fast Attention Through Optimizing Memory." On a high level, the concept that resonates most with me is your focus on bandwidth and data movement. It was interesting to see how much performance is lost simply because the data isn't "shaped" correctly for the hardware, and the performance gains when addressing those issues.

This has actually been the primary challenge in many of my recent projects. I've found that the bottleneck is not always raw computational power, but rather how fast I can move massive amounts of data through the

system. Notably when moving terabytes of high-resolution images for my photogrammetry workstation project.

Because of this, I am very interested in exploring:

- HPC in AI inferencing vs. training
- HPC in computer vision and physical AI
- HPC at the edge

Are there other emerging bottlenecks or specific hardware architectures you think would be worth investigating? I am excited to start diving into a project and would value any further direction you might suggest.

I look forward to hearing from you.

Thank you,

Aayan Sharma

[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Tue, Mar 10, 2026 at 7:03 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Dear Professor Low,

I hope you are doing well. I wanted to follow up on my previous email and see if you had any further thoughts.

I'm excited to keep exploring this area and would be happy to meet this week for 15 minutes to discuss further, if that would be helpful.

Thank you,
Aayan Sharma
[Quoted text hidden]

Tze Meng Low <lowt@andrew.cmu.edu> Wed, Mar 11, 2026 at 12:12 PM
To: Aayan Sharma <aayans@andrew.cmu.edu>

Hi Aayan,

We can definitely come up with a project that is suitable for you in the 3 areas you have mentioned. Let me get back to you on more details perhaps next week.

Best regards,
Tze Meng
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Thu, Mar 12, 2026 at 6:31 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Sounds great, thank you!
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Thu, Mar 26, 2026 at 11:10 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Dear Professor Low,

I hope you are doing well. I wanted to follow up and see if you had any further thoughts on a potential project for the summer.

I'm still very excited about the three areas we discussed, HPC in AI inferencing, computer vision, and edge computing, and am ready to dive in whenever you have direction for me.

On that note, I recently came across a paper on Tsetlin machines, a logic-based alternative to neural networks with some interesting energy efficiency properties. I thought it could be a potential point of research. What do you think? Below is the wikipedia article I came across.

https://en.wikipedia.org/wiki/Tsetlin_machine

Thank you,
Aayan Sharma
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Sun, Apr 5, 2026 at 3:23 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Hi Professor Low,

I hope you're doing well. I wanted to follow up about doing research with you this summer.

I'll be on campus from June 22nd to July 30th for Summer 2 classes, and I'd like to figure out a project before the May 8th SURA deadline. I'm looking forward to working with you, I'd be happy to meet this week or next week after Carnival if that works.

Thank you,
Aayan Sharma
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Mon, Apr 13, 2026 at 5:06 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Hi Professor Low,

I hope you're doing well. I wanted to follow up on our research project. As you know, I'm registering for classes over the summer and would like to work out some details around the project.

I've also been thinking more about the research direction, my dad works at Analog Devices in data center and energy, and it's given me a lot of context around edge processing trends. It's something I'm excited to explore further.

I do want to mention that I'm also in the process of sorting out housing for the summer, so I'm hoping to get the research details finalized soon so I can get everything moving in the right direction.

Would it be possible to swing by your office this week? Please let me know what times this week works for you.

Thank you,
Aayan Sharma
[Quoted text hidden]

Tze Meng Low <lowt@andrew.cmu.edu> Tue, Apr 14, 2026 at 12:59 PM
To: Aayan Sharma <aayans@andrew.cmu.edu>

Hi Aayan,

Sorry for the long silence. Would you be able to meet on this coming Friday morning? 4/24?

Best regards,
Tze Meng

[Quoted text hidden]

Tze Meng Low <lowt@andrew.cmu.edu> Tue, Apr 14, 2026 at 1:05 PM
To: Aayan Sharma <aayans@andrew.cmu.edu>

Sorry. I meant Apr 17.

Best regards,
Tze Meng
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Tue, Apr 14, 2026 at 2:18 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Hi Professor Low,

I am available from 1-6PM on Friday the 17th. Let me know what times work best for you.

I look forward to finalizing the project details.

Thank you,
Aayan Sharma
[Quoted text hidden]

Tze Meng Low <lowt@andrew.cmu.edu> Tue, Apr 14, 2026 at 2:43 PM
To: Aayan Sharma <aayans@andrew.cmu.edu>

How about 1pm?

Best regards,
Tze Meng
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Tue, Apr 14, 2026 at 3:31 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Yes, that works for me. Looking forward to it.
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Sat, Apr 18, 2026 at 11:13 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Hi Professor Low,

I hope you're doing well. It was great speaking with you yesterday, and I'm really excited about the opportunity.

I wanted to confirm the availability we discussed. From May 5th to June 19th, I can commit around 35 hours per week. Once I'm back in Pittsburgh from June 19th through July 30th, that drops to roughly 20 hours per week due to a summer course I'm taking. From early August to the first day of classes, I can commit 35 hours per week.

Please let me know what the next steps look like. I'm looking forward to getting started.

Thank you,
Aayan Sharma
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Sun, May 3, 2026 at 7:07 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Hi Professor Low,

I hope you are doing well. I wanted to follow up regarding logistics for research. If you could let me know what the next steps are that would be great!

Thank you,
Aayan Sharma
[Quoted text hidden]

Tze Meng Low <lowt@andrew.cmu.edu> Wed, May 6, 2026 at 10:56 AM
To: Aayan Sharma <aayans@andrew.cmu.edu>

Hi Aayan,

There's really nothing to do on your part. For you, once you have recovered from Finals, then it's really the tasks that we have talked about. I'm working with the admin side to get you officially hired.

Best regards,
Tze Meng
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Wed, May 6, 2026 at 8:19 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Hi Professor Low,

Thank you for the update.

I've finished finals and I'm happy to start working on Monday. I wanted to see if we could find a time for a quick call before then to make sure I'm aligned on where to focus first and what to work on next.

Please let me know if there's a time that works for you.

Thank you,
Aayan Sharma
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Fri, May 15, 2026 at 12:12 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Hi Professor Low,

I wanted to follow up on my previous email. I'm really excited to get started on the research and want to make sure I'm hitting the ground running from day one.

Would it be possible to find a time to meet or chat briefly so I can make sure I'm focused on the right things going in?

Thank you,
Aayan Sharma
[Quoted text hidden]

Tze Meng Low <lowt@andrew.cmu.edu> Thu, May 21, 2026 at 1:31 PM
To: Aayan Sharma <aayans@andrew.cmu.edu>

Hi Aayan,

Sorry for being silent for so long. How about meeting on the upcoming Tuesday after the long weekend?

Let me know if what time works for you.

Best regards,
Tze Meng
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Thu, May 21, 2026 at 1:40 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Yes, that works. I'm available all day via zoom or Google meet. Let me know what time works best.
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Thu, May 21, 2026 at 1:43 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Oh sorry, does 11AM work for you?
[Quoted text hidden]

Tze Meng Low <lowt@andrew.cmu.edu> Tue, May 26, 2026 at 10:01 AM
To: Aayan Sharma <aayans@andrew.cmu.edu>

That works. Let's meet on my zoom. https://cmu.zoom.us/my/tzemeng.

See you at 11am.
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Tue, May 26, 2026 at 10:19 AM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Sounds good. Unfortunately, some stuff has come up and I'll be able to join at around 11:20AM. Thank you for your understanding.
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Tue, May 26, 2026 at 10:59 AM
To: Tze Meng Low <lowt@andrew.cmu.edu>

I'm in the meeting. Ready whenever you are.
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Wed, May 27, 2026 at 2:29 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Hi Professor Low,

I wanted to give a quick update on my progress so far. I've implemented and benchmarked Gaussian blur across multiple kernel sizes, and I've also implemented a Sobel filter as part of the OpenCV benchmarking portion of the project. I've been using Claude heavily for code generation, iteration, and optimization ideas while working through different convolution implementations and benchmarking setups.

I was wondering if there are any resources available through CMU or through the research project for additional Claude/API usage. My personal Claude subscription has started becoming a limiting factor given the amount of experimentation and iteration involved.

I also came across information about CMU's AI Gateway and was curious whether that might be relevant for this project, or whether there are other research credits/resources that students typically use for this kind of work.

Thank you, and I'll continue preparing more benchmarking results for our next meeting.

Thank you,
Aayan Sharma
[Quoted text hidden]

Tze Meng Low <lowt@andrew.cmu.edu> Thu, May 28, 2026 at 11:10 AM
To: Aayan Sharma <aayans@andrew.cmu.edu>

Hi Aayan,

It sounds like you already did a lot. Let's discuss what you have done, and do some consolidation before we move forward. We can definitely get more credits.

Best regards,
Tze Meng

On Wed, May 27, 2026 at 2:29 PM Aayan Sharma <aayans@andrew.cmu.edu> wrote:

> Hi Professor Low,
>
> [Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Thu, May 28, 2026 at 4:45 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Hi Professor Low,

Thank you for your response, here's a brief update on what I've been working on.

I benchmarked OpenCV's Gaussian blur (cv2.GaussianBlur, optimized separable path) and Sobel filter on a 4K image (3840×2160) across 8 kernel sizes (7×7 to 101×101), averaging 100 runs each to establish a baseline. I then had AI generate a C implementation of Gaussian blur and compared it against that baseline. The unoptimized C version ran ~29–39x slower than OpenCV. After applying a first optimization (detailed in the attached spreadsheet), the gap narrowed to ~15–32x, roughly a 2x improvement in the C implementation itself. It's interesting to note how the 15x15 kernel is an outlier in performance.

Would you have time to meet Friday or Monday to discuss next steps and the additional credits? I will continue optimizing to get OpenCV level performance.

Thank you,
Aayan Sharma
[Quoted text hidden]

Results.xlsx
26K

Tze Meng Low <lowt@andrew.cmu.edu> Mon, Jun 1, 2026 at 11:15 AM
To: Aayan Sharma <aayans@andrew.cmu.edu>

There seemed to be some problem with my email. I just got this message today.

Let's meet at our usual slot on Tuesday. We may need the entire hour to discuss. Please prepare to show code as well.

Best regards,
Tze Meng
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Mon, Jun 1, 2026 at 11:16 AM

To: Tze Meng Low <lowt@andrew.cmu.edu>

Sure, no problem.
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Tue, Jun 2, 2026 at 10:42 AM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Attached is my code and results for reference for our meeting today.
[Quoted text hidden]

4 attachments

Results.xlsx
49K

benchmark_c_vs_opencv.py
4K

gaussianbluroptimization.c
15K

Makefile
1K

Aayan Sharma <aayans@andrew.cmu.edu> Wed, Jun 10, 2026 at 10:52 AM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Hi Professor Low,

I hope you are doing well. I wanted to reach out regarding the SIMD implementation of the Gaussian blur.

The main idea I've been exploring is that each output pixel in a row is independent of every other output pixel and no pixel needs another's result to compute its own. This means you can apply the same weight to many pixels simultaneously, since they are all doing the exact same operation on different data. By thinking about it this way, the same instruction can be applied across many pixels at once, which is where SIMD can be exposed. The loop restructuring follows naturally from this idea. I've attached my thought process/implementation to this email.

I wanted to check in to see if I am on the right track. I am happy to schedule another meeting to discuss next steps whenever it works for you.

Thank you,
Aayan Sharma
[Quoted text hidden]

SIMD Gaussian Blur Implementation.pdf
134K

Tze Meng Low <lowt@andrew.cmu.edu> Thu, Jun 11, 2026 at 11:07 AM
To: Aayan Sharma <aayans@andrew.cmu.edu>

Hi Aayan,

You are on the right track. Now, some questions you need to answer are
1. How many pixels should you compute at the same time,
2. Assuming you have a small function that computes the gaussian blur for a small number of pixels, how do you use it to compute the entire gaussian blur (excluding edge cases).

Best regards,
Tze Meng
[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Fri, Jun 12, 2026 at 1:13 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Hi Professor Low,

Thank you for the questions. Here are my answers:

1. A NEON register is 128 bits wide. Since an integer is 32 bits, we can fit 4 values per register. So we compute 4 pixels at a time.

2. We fully update all 4 pixels before moving to the next group. The function slides across the row in steps of 4, computing the complete output for each group before moving on.

Thank you,
Aayan Sharma
[Quoted text hidden]

Tze Meng Low <lowt@andrew.cmu.edu> Fri, Jun 12, 2026 at 3:00 PM
To: Aayan Sharma <aayans@andrew.cmu.edu>

Hi Aayan,

The answers worked out mathematically. However, there's a couple of things to consider:

1. Is there a need to fully update all 4 pixels before moving to another group of 4 pixels? Consider the number of registers available on an ARM processor.

2. Do you want to compute the entire row before computing the next row? Consider what inputs are required to compute each output pixel.

Other considerations:

- Do we still want to use integers? Why not floats since they are also 32bits?
- If we do use integers, do we need 32bits for integers? Can we get away with lesser number of bits?
- Are there the appropriate SIMD instructions on the ARM processor?

Best regards,
Tze Meng

[Quoted text hidden]

Aayan Sharma <aayans@andrew.cmu.edu> Mon, Jun 15, 2026 at 1:10 PM
To: Tze Meng Low <lowt@andrew.cmu.edu>

Hi Professor Low,

Here are the answers to each question:

1) Is there a need to fully update all 4 pixels before moving to another group of 4 pixels?

No. This is because each group of pixels has its own accumulation, and each group of pixels is independent of the next group of pixels. With 32 SIMD/NEON registers, we can compute multiple groups of pixels at once. Ideally, we can compute 128 pixels at once, but in practice we won't be able to use all 32 registers.

2) Do you want to compute the entire row before computing the next row?

Rows are independent in the same way pixels are, output row r+1 does not depend on output row r's result, it is only dependent on the input rows. However, the input rows overlap heavily. So processing rows in order lets us re-use a lot of the input data in cache rather than re fetching it if we were to compute each row one at a time. This seems to be more of a memory movement optimization.

Do we still want to use integers? Why not floats since they are also 32 bits?

A float is also 32 bits, so it would give the same number of lanes per register as int32. But our pixel values and accumulators don't need floating point precision, and using floats would require converting 8 bit pixel values to a float and back at the end which adds extra instructions.

If we do use integers, do we need 32 bits? Can we get away with fewer bits?

The idea of the code I've written is that we don't need 32 bits. The kernel is normalized to a sum of 256 (8 bit pixel values), so the worst case accumulator value is at most 256*255. = 65280, which fits within the range of a 16 bit integer value. So we can double the number of pixels per SIMD instruction (8 vs 4)

Are there appropriate SIMD instructions on the ARM processor?

Yes, NEON has widening multiply-accumulate instructions (vmlal_u8) that take a 8 bit integer input and accumulate directly into a 16 bit register in a single instruction, which matches our exact data types.

Let me know if this reasoning makes sense, or if there's anything I should reconsider.

Thank you,
Aayan sharma
[Quoted text hidden]